

Alain Ignacio

Agent Infrastructure Engineer | Founder, Isagawa

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github.com/isagawa-co (kernel, specs) · github.com/isagawa-qa (QA platforms)

Portfolio: isagawa.co

Summary

I build agent infrastructure. The core loop is always the same: the agent starts, anchors to its protocol, works under runtime enforcement, learns from failures, and completes. That loop governs every agent I ship. I designed and built the Isagawa Kernel, an open-source agent runtime where AI agents construct their own governance infrastructure from authored specifications. The system enforces guardrails at the tool-call boundary, not in the prompt. Twenty years of quality systems engineering across healthcare, government ERP, telecom, and startups led directly to this work. The discipline is identical: find where processes break, build mechanical prevention so they cannot break that way again. First applied to software testing, now applied to governing autonomous agents in production.

Isagawa: Founder & Agent Infrastructure Architect

2025 to Present · isagawa.co

80+ attested pipeline runs · 800+ autonomous tasks executed · 9 production repos · All signed to Rekor

Isagawa Kernel (Open Source, MIT) · github.com/isagawa-co/isagawa-kernel

An agent runtime and governance system. The agent scans any repository, writes its own protocol and enforcement hooks, then improves them mechanically every time it fails. The core loop: session-start, anchor, work, learn, complete. Every session. Every agent.

When a test fails or a protocol violation is detected, the agent is blocked from writing until it records a lesson. That lesson updates both the protocol and the enforcement rules simultaneously. Hooks run at the tool-call boundary, making guardrail bypass architecturally impossible. A re-centering mechanism fires every N actions, forcing protocol re-read and accumulated lesson review. Drift cannot happen over long sessions because the loop prevents it. State survives context compaction and restarts. The agent resumes exactly where it left off.

The kernel is domain-agnostic agent infrastructure. It governs how the agent works. Pair it with an agent harness to define what it builds.

Multi-Agent Orchestration System (Proprietary) · github.com/isagawa-co/domain-spec-factory

Autonomous construction of governed agent harnesses. Given an industry vertical as input, a kernel-governed agent researches the domain, builds a complete agent harness from original research, and validates it before publishing. The kernel governs the agent that builds kernel-governed harnesses. 14 production agent harnesses manufactured autonomously across healthcare, DevOps, compliance, real estate, and financial verticals.

The emerging SDD ecosystem (Amazon Kiro, GitHub Spec Kit, BMAD) produces planning artifacts that guide agents. None enforce guardrails during execution or learn from failures after. This system closes both gaps. Each harness is an operational artifact that self-assembles a governed workspace with runtime enforcement and quality gates built in.

Agentic Workflow Pipeline

Intent capture, atomic task decomposition with gate contracts, headless batch execution (one agent instance per task), completion validation, and cryptographic attestation. One sentence of intent becomes a verified, signed artifact with no human intervention. 80+ signed runs. The pipeline built the portfolio site that describes it.

Agent Quality Platforms (Proprietary)

github.com/isagawa-qa · 5 production platforms on the Isagawa agent runtime

Five agent-driven quality automation platforms sharing the same pattern: a kernel-governed agent reads reference

implementations, enforces layered architecture during generation, and improves permanently after every failure. Each platform is itself a multi-agent system operating under runtime governance.

LLM Evaluation Platform (DeepEval). Agent-driven evaluation across 29+ metrics covering RAG, Chat, Agent, and Conversational pipelines. The orchestrating agent selects metrics, sets thresholds from tested defaults, generates pytest-native eval suites, and runs them with failure triage. Golden datasets drive every evaluation.

Playwright Automation Platform. Agent-governed TypeScript/Playwright test generation with three paths (UI, API, hybrid) sharing layered architecture. Playwright MCP provides real-time element and endpoint discovery during generation. Strict separation: locators, endpoints, business logic, and assertions each live in their designated layer.

Selenium Automation Platform. Same governed architecture in Python/Selenium/pytest. The agent discovers page elements, generates Page Objects, Tasks, Roles, and Tests following strict architectural separation. Runtime enforcement prevents layer mixing and pattern drift across sessions.

SSH Compliance Platform · isagawa.co/ssh-compliance. Automated compliance validation across 8 security frameworks (STIG, CIS, NIST, FIPS, PCI DSS, HIPAA, SOC 2, ISO 27001). Connects to infrastructure, executes every check, produces auditor-ready reports with captured command output as evidence.

Docker Compliance Platform. Container image validation against CIS, DISA-STIG, and FIPS standards. Pulls the image, detects OS and package manager, runs every check, produces structured compliance reports per rule, per framework, per image.

Also Shipped (isagawa-co, Proprietary)

Vibe Coder · github.com/isagawa-co/vibe-coder-spec. Agent-driven application building for non-technical founders. The kernel-governed agent handles all technical decisions and explains every choice in plain English. The same agent governance model applied to a consumer product direction.

AutoApply · isagawa.co/job-application. End-to-end job application pipeline using Playwright MCP for universal form discovery, profile-driven form fill across any ATS, and a hook-enforced human review gate before submission.

Attestation Pipeline · isagawa.co/attestation. Every pipeline run signed with Sigstore and logged to Rekor. The full intent chain from hashed user input through shipped artifact is verifiable on a public tamper-evident log.

Prior Experience: Quality Systems Engineering

QA Lead · HMSA · March 2026 to Present (Remote)

Returned to Hawaii's largest health insurer to lead QA for claims processing and stand up a new automation function. Testing scope covers adjudication workflows and EDI integrations across UI, API, and database layers within enterprise compliance constraints. The same systems thinking that drives agent governance applied to healthcare quality assurance.

Senior QA Engineer · Helios Digital · May 2025 to March 2026 (Remote)

Founding QA hire for a seed-stage AI financial analysis platform. Built a QA Center of Excellence achieving 4-5x improvement in test velocity per engineer. Reduced new engineer onboarding from three weeks to one week through enforcement patterns that agents and humans both follow. Integrated Claude Code into daily development workflows.

Senior QA Automation Engineer · Nakupuna Consulting · Nov 2022 to May 2025 (Remote, Navy contract)

Established the test automation function from scratch for Navy PeopleSoft ERP in a highly restricted environment. Scaled the team from zero to four engineers within six months, getting Python/Selenium approved where only legacy tools had been authorized. Designed a framework architecture that the development team adopted over native tooling for their own testing.

Test Manager · HMSA · May 2013 to Nov 2022 (Honolulu, HI)

Nine years at Hawaii's largest health insurer progressing from QA Analyst to Test Manager. Defined QA strategy and testing standards across Claims Processing, EDI, Membership, Billing, Enrollment, HL7-FHIR, and RESTful services. Built API automation and end-to-end test frameworks.

Senior Test Analyst · IBM · July 2008 to May 2013 (Honolulu, HI)

Led QA for Virgin Mobile USA's billing platform across website, CRM, WAP, and charging/rating systems. Managed an 18-member onshore/offshore team across 15+ enterprise releases.

QA Lead · Virgin Mobile USA · 2004 to 2008 (Walnut Creek, CA)

Promoted from tester to lead for the billing platform, overseeing testing for MVNO pay-as-you-go services.

QA Lead · Sony Electronics · 1999 to 2004 (San Jose, CA)

Promoted from tester to lead for handheld devices and VAIO product lines.

Technical Skills

Agent Infrastructure: agent runtime design, multi-agent orchestration, loop-based governance architecture, tool-call interception, guardrails and runtime enforcement, agent observability, session persistence, context management, MCP server design, Playwright MCP, Claude Code

Agent Evaluation: DeepEval, RAG evaluation, agent evaluation, golden dataset design, pytest-native eval suites, 29+ metric frameworks, agent quality measurement

Architecture: Spec-Driven Development, agentic workflow design, agent protocol design, defense-in-depth enforcement, state machines, event-driven workflows, domain-driven design

Test Automation: Playwright, Selenium, pytest, Cypress, layered framework architecture, Page Object Model, BDD

Compliance and Security: STIG, CIS, NIST, FIPS, PCI DSS, HIPAA, SOC 2, ISO 27001, Sigstore, Rekor, cryptographic attestation

Languages and Tools: Python, TypeScript, JavaScript, SQL, Git, FastAPI, Postman